Exact Word Replacement and Its Coverage Boundary

Collatz proof project

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Abstract

We formalize an exact merging criterion for finite parity words, including the classical Garner stem family. We also prove that the convergent seed 27 has no smaller positive merging predecessor with both equal time and equal odd count, at any time. The results validate a local replacement method but refute its unrestricted universal coverage. They do not settle coverage for a hypothetical least unbounded seed or prove Collatz. We also bound lower-count replacements and test varying odd counts through depth 18, accounting for every cylinder quotient.

Use $T(n) = n/2$ for even n, and $(3n+1)/2$ for odd n. For a word w, let $|w|$ be its length, $j(w)$ its number of ones, and $D(\varnothing) = 0$, $D(bw) = b3^{j(w)} + 2D(w)$, where $b \in \{0,1\}$. When n realizes w, the exact identity is $2^{|w|}T^{|w|}(n) = 3^{j(w)}n + D(w)$.

Theorem 1 (Exact replacement)**.** *Suppose n realizes u, $|u| = |v| = t$, and $j(u) = j(v) = j$. A natural x realizes v and satisfies $T^t(x) = T^t(n)$ if and only if*

$$3^j x + D(v) = 3^j n + D(u).$$

In particular, if $D(v) = D(u) + 3^j\delta$ and $0 < \delta < n$, then $x = n - \delta$ is a smaller positive merging predecessor.

Proof. Necessity follows from the two affine identities. Conversely, the displayed equality makes $3^j x + D(v)$ divisible by 2^t. The repository's `realizes_iff_dvd` theorem proves every prescribed parity from this divisibility. The affine identity then determines the endpoint uniquely. ☐

Theorem 2 (Classical Garner stems)**.** *For $a \geq 1$, set $u = 1^a00$ and $v = 01^{a-1}01$. Every $n > 1$ realizing u merges with $n-1$, realizing v, after $a+2$ steps.*

Proof. Both words have length $a+2$ and odd count a. Their corrections are $D(u) = 3^a - 2^a$ and $D(v) = 2 \cdot 3^a - 2^a$. Apply the first theorem with $\delta = 1$. ☐

This family is classical, not a novelty claim; see Elia and Tucker, Section 3[1]. For example, 15 and 14 reach 20 after six steps with four odd steps each; all prefix coefficients of the six-step word from 15 are at least one.

Theorem 3 (All-time coverage boundary)**.** *For every $0 < x < 27$ and every $t \geq 0$,*

$$\neg\big(T^t(x) = T^t(27) \text{ and } j_t(x) = j_t(27)\big).$$

Proof. Lean's kernel checks all $x = 1, \ldots, 26$ and $t = 0, \ldots, 70$. At time 70, it also checks $T^{70}(27) = 1$, $j_{70}(27) = 41$, while $T^{70}(x) \in \{1, 2\}$ and $j_{70}(x) \leq 35$ for all those x. Induction on k proves $j_k(2) \leq j_k(1) \leq j_k(2)+1$. The odd-count cocycle therefore gives $j_{70+k}(x) \leq 35 + j_k(1) < 41 + j_k(1) = j_{70+k}(27)$. ☐

[1] `https://arxiv.org/abs/1511.09141`

Allowing the odd count to change

Theorem 4 (Lower-count size bound)**.** *If $T^t(x) = T^t(n)$ and $j_t(x) < j_t(n)$, then*

$$3n < x + 2^{t-j_t(x)}.$$

Consequently, if $2^t \le n$, then $2n < x$.

Proof. Write $a = j_t(x)$ and $b = j_t(n)$. The exact form and the universal shifted-height bound give

$$3^{a+1}n \le 3^b n \le 2^t T^t(n) < 2^t(T^t(x) + 1) \le 3^a(x + 2^{t-a}).$$

Cancel 3^a. If $2^t \le n$, then $2^{t-a} \le n$, giving $2n < x$. Both conclusions are formalized in Lean. $\square$

The full quotient test. Put $L = 2^t$ and choose canonical seeds $0 \le r, s < L$ with odd counts j, k and endpoints e, f. The library proves $e < 3^j$, $f < 3^k$ and

$$T^t(r + Lq) = e + 3^j q, \qquad T^t(s + Lp) = f + 3^k p.$$

Thus an exact merge is equivalent to $f + 3^k p = e + 3^j q$. When $k < j$, a smaller merging predecessor can occur only for $q = 0$, by the preceding theorem. In that case $s + Lp < r < L$ forces $p = 0$, so it suffices to test $f = e$ and $0 < s < r$.

When $k > j$, compatibility is exactly $f \equiv e \pmod{3^j}$. Write $f = e + 3^j q_0$ and $d = k - j > 0$. All compatible nonnegative quotients are

$$q = q_0 + 3^d Q, \qquad p = Q, \qquad Q \ge 0.$$

The resulting endpoint equality is `higher_count_merge_lifts`. For $Q \ge 1$, the replacement is positive and smaller because

$$(r + Lq) - (s + LQ) = r - s + L\big(q_0 + (3^d - 1)Q\big) > 0.$$

Consequently a finite residue intersection detects infinitely many useful lifts; testing only the canonical seed would miss them.

Finite evidence and remaining obligation. An exhaustive Python census at depth 18 certifies replacements for 193,309 of 262,144 binary cylinders, including 1,391 of the 7,495 whose every prefix coefficient is at least one. Each reported canonical merge is independently replayed; equal counts extend it to the whole cylinder. These counts are experimental, not kernel-certified coverage. The new all-quotient test finds no additional coverage from different odd counts among the remaining 6,104 noncontracting-prefix cylinders at depth 18. Both tests also give zero additions at every depth 1 through 17. This is a finite-depth observation; no theorem of redundancy at arbitrary depth has been proved. A smaller positive merging predecessor contradicts least-unboundedness; that implication is formalized without imposing noncontraction on its path. The missing theorem is existence for every hypothetical least unbounded seed. The counterexample 27 is convergent and does not refute that conditional statement. Variable lengths or odd counts remain possible.

Verification. The eight public declarations in `Collatz.WordSurgery` are included in the axiom audit. The census and independent small-depth tests are `analysis/word_surgery.py` and `test_word_surgery.py`. The variable-count comparison is `variable_count_surgery.py`, with a separate test file. Canonical rows are independently replayed through depth 9 and the quotient formulas on all word pairs through depth 6. No unproved Collatz premise is discharged by the finite computation.